DO NOW — RECALL

Quicksort: pivot and partition

Name: _____ Class: _____ Date: _____

1. In bubble sort, which value reaches its final position at the end of pass one?

2. What is a base case?

3. Which element is a[3] in [6,2,9,4,5]?

4. What is the result of 4 <= 6 and why?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The largest value in the unsorted section.

2. Model answer

A condition that stops recursive calls.

3. Model answer

4; indexes begin at 0.

4. Model answer

True, because 4 is less than or equal to 6.

Vocabulary — know these terms

pivot — Element selected to split a quicksort partition.

partition — Rearrange values so smaller/equal values are on one side of the pivot.

Lomuto — A partition approach scanning left to right and placing the pivot last.

index — Zero-based position within the array.

divide and conquer — Solve smaller subproblems and combine results.

NEW LEARNING

Read, annotate and explain

Pick a pivot

Use the rightmost element, 6. A left boundary i tracks where the next small value belongs.

Scan and move

Only values less than or equal to 6 move left. We move 2, then 4, without sorting those two values.

Place the pivot

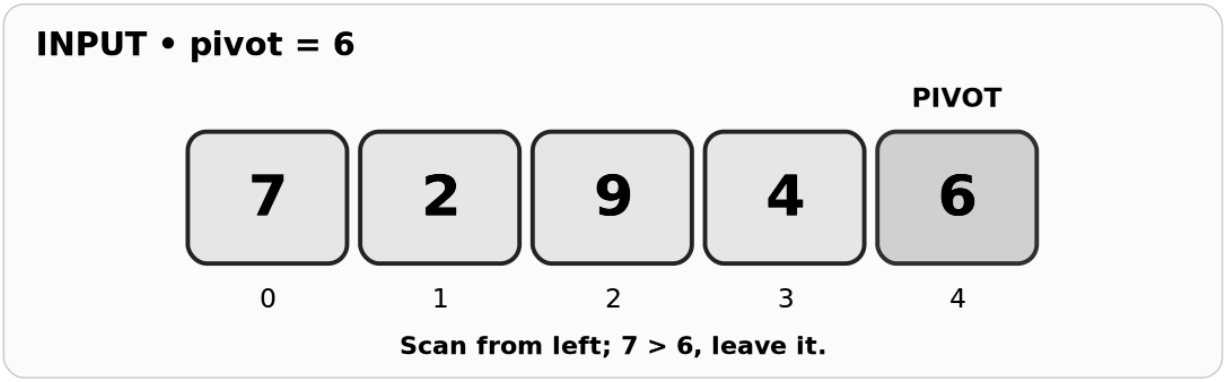
Swap 6 into index 2; elements to the left are <=6, and those to the right are >6. Each side still needs sorting.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

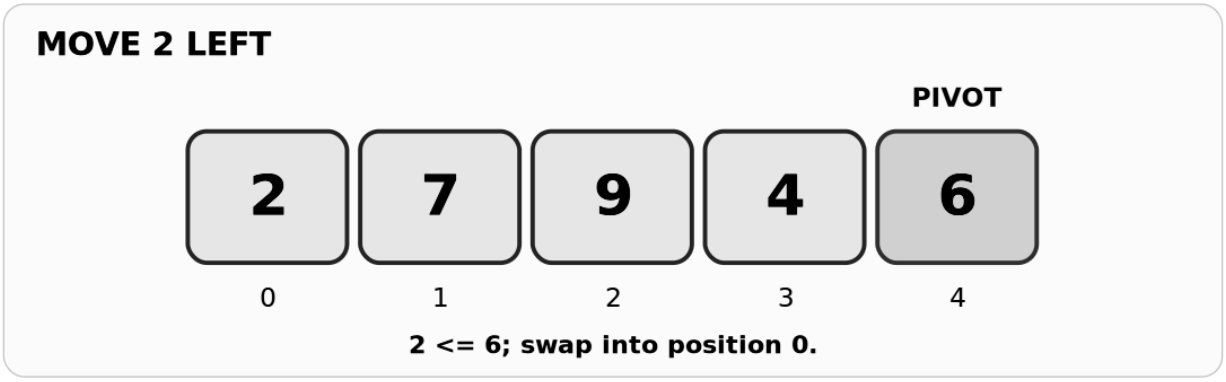
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INPUT • pivot = 6: 7, 2, 9, 4, 6. Scan from left; 7 > 6, leave it.

New Learning

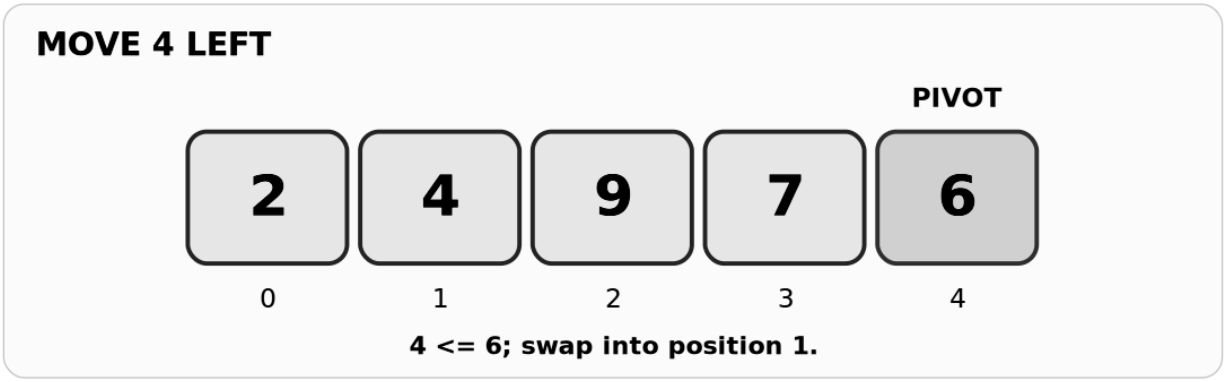


MOVE 2 LEFT: 2, 7, 9, 4, 6. 2 <= 6; swap into position 0.

New Learning · original lesson diagrams

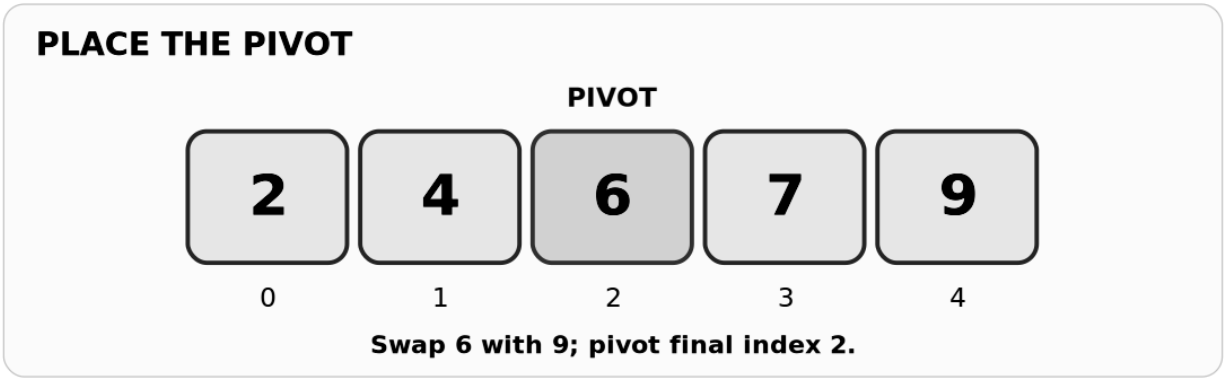
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



MOVE 4 LEFT: 2, 4, 9, 7, 6. 4 <= 6; swap into position 1.

New Learning



PLACE THE PIVOT: 2, 4, 6, 7, 9. Swap 6 with 9; pivot final index 2.

I DO — WORKED EXAMPLE

Follow the worked method

Partition [6,3,8,2,5] around final-element pivot 5.

Predict the next state

6 is larger than 5; 3 is smaller and swaps into the left boundary.

Reveal and explain

8 is larger; 2 swaps to index 1, giving [3,2,8,6,5].

Reveal and explain

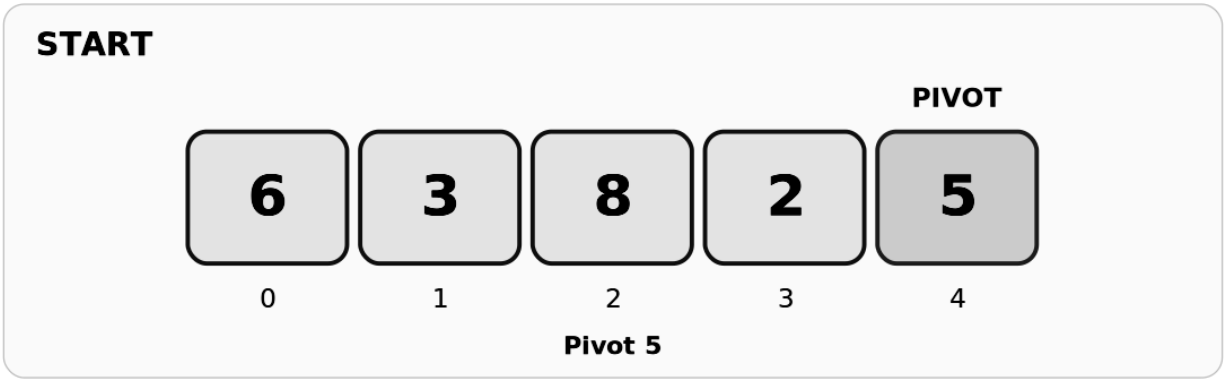
Swap pivot 5 with position 2: [3,2,5,6,8]. The two sides are not yet necessarily sorted.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [6,3,8,2,5] around final-element pivot 5.

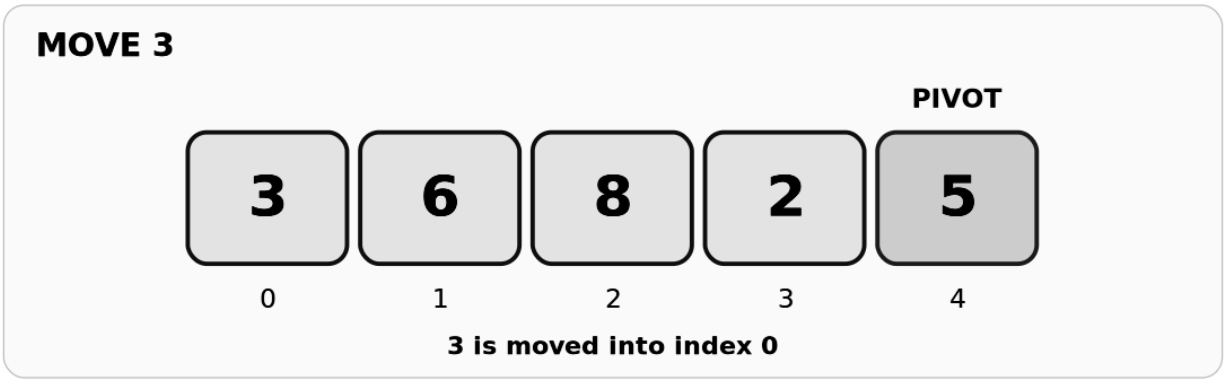
6 is larger than 5; 3 is smaller and swaps into the left boundary.



START: 6, 3, 8, 2, 5. Pivot 5

Partition [6,3,8,2,5] around final-element pivot 5.

6 is larger than 5; 3 is smaller and swaps into the left boundary.



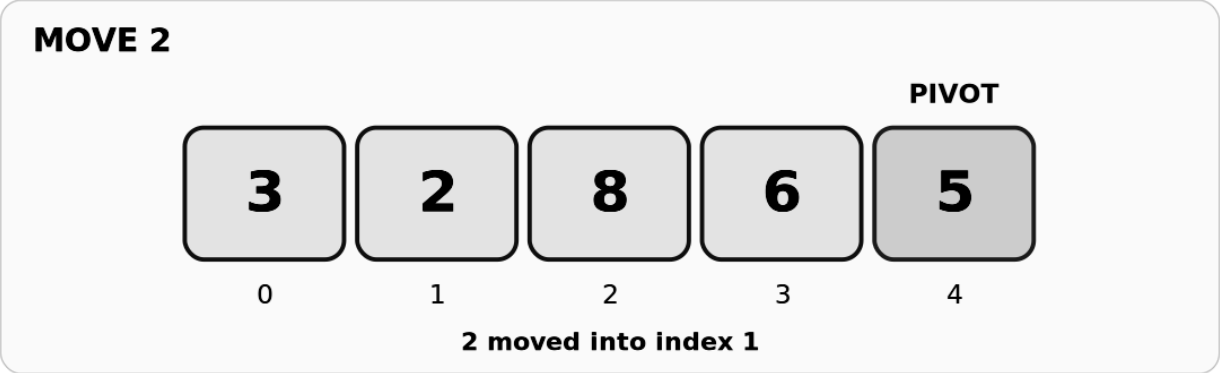
MOVE 3: 3, 6, 8, 2, 5. 3 is moved into index 0

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [6,3,8,2,5] around final-element pivot 5.

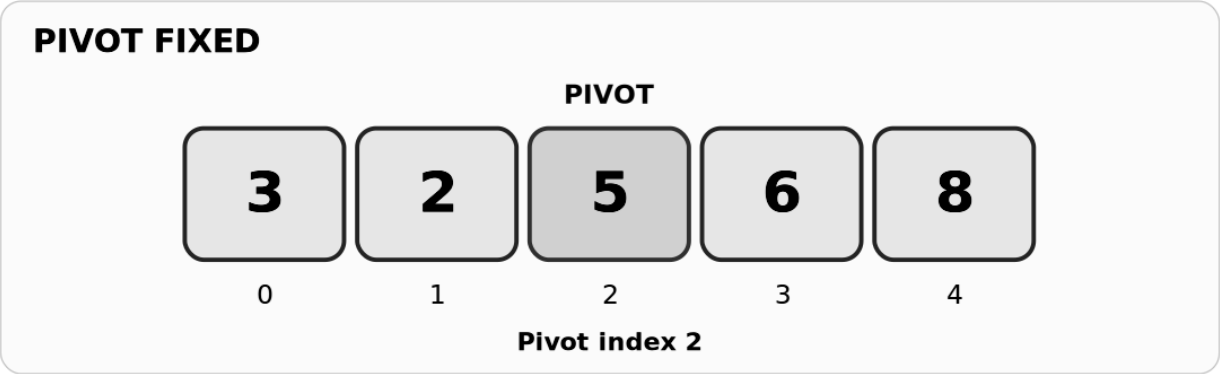
6 is larger than 5; 3 is smaller and swaps into the left boundary.



MOVE 2: 3, 2, 8, 6, 5. 2 moved into index 1

Partition [6,3,8,2,5] around final-element pivot 5.

6 is larger than 5; 3 is smaller and swaps into the left boundary.



PIVOT FIXED: 3, 2, 5, 6, 8. Pivot index 2

WE DO — GUIDED PRACTICE

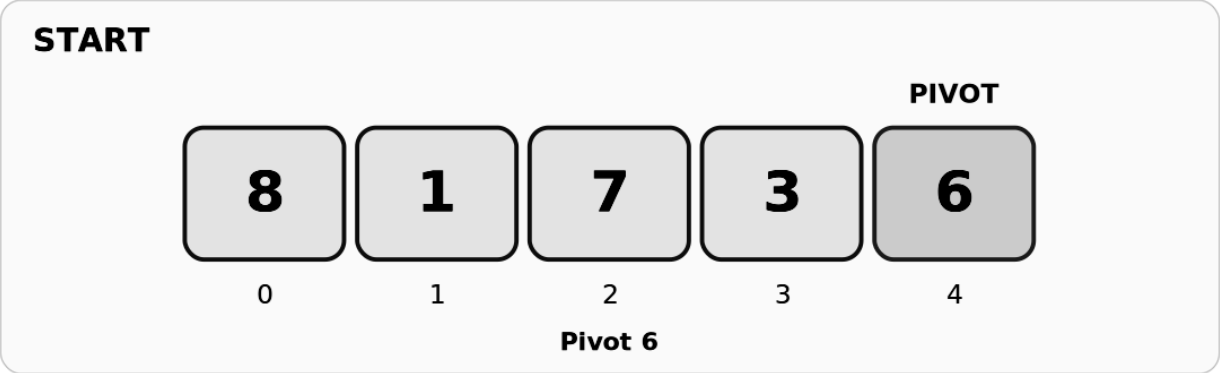
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [8,1,7,3,6] using final pivot 6.

Compare 8 and 1 with pivot 6. Move 1 into index 0.



START: 8, 1, 7, 3, 6. Pivot 6

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the next state Compare 8 and 1 with pivot 6. Move 1 into index 0.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Partition [4,1,3] using the final value as pivot, with Lomuto partition. Give the result and pivot index.

Question 2

Partition [9,2,5,1,6] using the final pivot. Show the array after moving each qualifying value and after placing the pivot.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why choosing the last element as pivot can lead to $O(n^2)$ work on already sorted data, and suggest an improvement.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A quicksort partition automatically fully sorts the two sides of the pivot.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

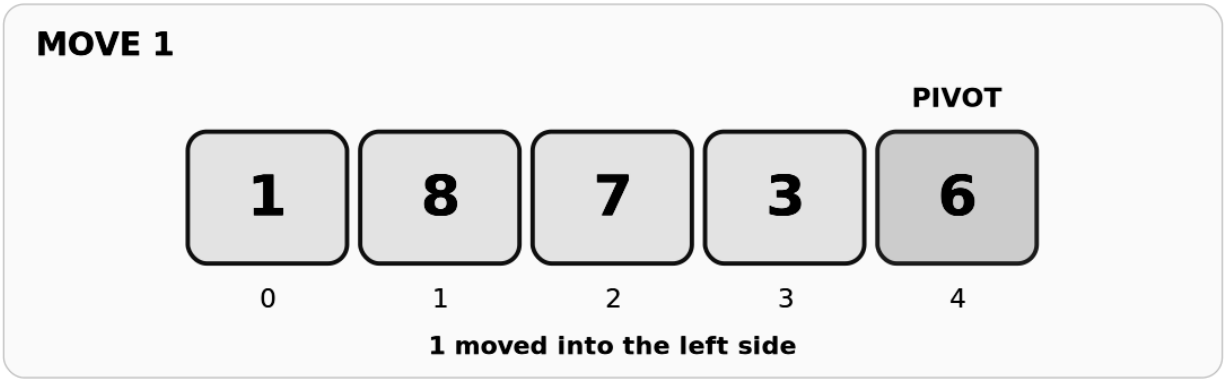
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [8,1,7,3,6] using final pivot 6.

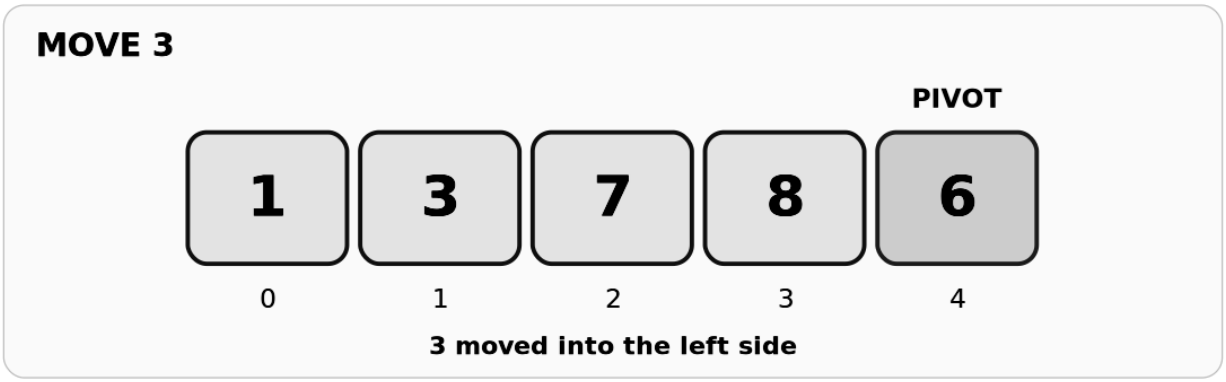
Compare 8 and 1 with pivot 6. Move 1 into index 0.



MOVE 1: 1, 8, 7, 3, 6. 1 moved into the left side

Partition [8,1,7,3,6] using final pivot 6.

Compare 8 and 1 with pivot 6. Move 1 into index 0.



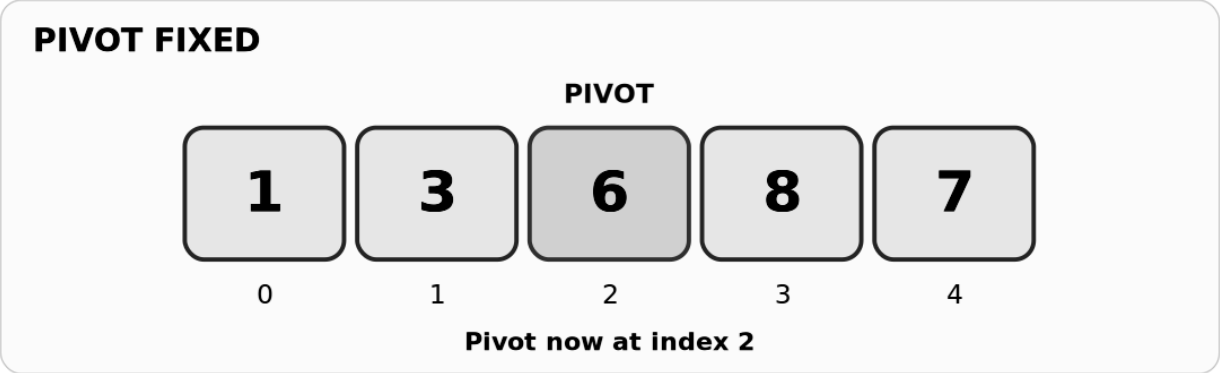
MOVE 3: 1, 3, 7, 8, 6. 3 moved into the left side

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [8,1,7,3,6] using final pivot 6.

Compare 8 and 1 with pivot 6. Move 1 into index 0.



PIVOT FIXED: 1, 3, 6, 8, 7. Pivot now at index 2

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Reveal and explain

7 is too large, then 3 moves to index 1: [1,3,7,8,6].

Put pivot at index 2: [1,3,6,8,7]. The right side [8,7] still needs sorting.

IMPROVE: Explain one correction to your own reasoning.